

TB262826U

Reg. No :
Name :

BACHELOR'S DEGREE (C.B.C.S) EXAMINATION, FEBRUARY / MARCH 2026
2025 ADMISSIONS REGULAR
SEMESTER II - SKILL DEVELOPMENT COMPONENT (SDC) (SOFTWARE DEVELOPMENT)
VSD2SDC04B25 - Operating Systems

Time : 2 Hours

Maximum Marks : 70

Part A

Answer all questions from the bunch of CO1. Each question carries 2 marks.

(3x2=6)

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|--|-------------------|
| 1. Define Batch Processing in the context of operating system evolution. | [CO1, Understand] |
| 2. Define OS. List two examples of Mobile OS. | [CO1, Remember] |
| 3. Define an Operating System. | [CO1, Remember] |

Part A

Answer all questions from the bunch of CO2. Each question carries 2 marks.

(3x2=6)

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| 4. Define Interprocess Communication (IPC). | [CO2, Understand] |
| 5. Define a Process in an Operating System. | [CO2, Remember] |
| 6. Differentiate between Preemptive and Non-Preemptive Scheduling. | [CO2, Understand] |

Part A

Answer all questions from the bunch of CO3. Each question carries 2 marks.

(3x2=6)

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| 7. Define Internal Fragmentation. | [CO3, Understand] |
| 8. Differentiate between fixed partition allocation and variable partition allocation . | [CO3, Understand] |
| 9. Define Swapping. | [CO3, Remember] |

Part A

Answer all questions from the bunch of CO4. Each question carries 2 marks.

(3x2=6)

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| 10. Define Sequential Access method. | [CO4, Understand] |
| 11. Define Thrashing in an operating system. | [CO4, Remember] |
| 12. Define Virtual Memory and state its primary benefit. | [CO4, Remember] |

Part B

Answer any 2 questions from the bunch of CO1. Each question carries 6 marks.

(2x6=12)

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|---|-------------------|
| 13. Explain the role of the Kernel and list the three types of Kernels. | [CO1, Understand] |
| 14. Explain the Microkernel Operating System structure with a diagram. List its advantages. | [CO1, Understand] |
| 15. Explain the various Services provided by the Operating System. | [CO1, Remember] |

Part B

Answer any 2 questions from the bunch of CO2. Each question carries 6 marks.

(2x6=12)

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| 16. Describe the different states of a process using a state diagram. | [CO2, Remember] |
| 17. Explain the Process Control Block (PCB). | [CO2, Understand] |
| 18. Explain the execution of a System Call with a diagram. | [CO2, Understand] |

Part B

Answer any 2 questions from the bunch of CO4. Each question carries 6 marks.

(2x6=12)

19. Discuss the Acyclic Graph Directory structure with neat diagram.

[CO4, Understand]

20. Explain the Page-Fault Frequency (PFF) strategy to control Thrashing.

[CO4, Understand]

21. List and explain the fundamental File Operations.

[CO4, Remember]

Part C

Answer any 1 question from the bunch of CO3. Each question carries 10 marks.

(1x10=10)

22. Discuss Segmentation as a memory management scheme with neat diagram and explain its advantages over Paging.

[CO3, Understand]

23. Explain Contiguous Memory Allocation, contrasting Fixed Partitioning and Variable Partitioning, and discuss Fragmentation.

[CO3, Understand]